

Muhammad Haroon

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PERSONAL SUMMARY

AI Engineering student who builds and ships end-to-end ML systems: a RAG-based customer support chatbot (FAISS + LLM APIs, containerized with Docker), a visual search engine reaching mAP 0.93 (live on Hugging Face Spaces), and CNN classifiers. Strong foundation in PyTorch, retrieval systems, and rigorous model evaluation, with deployment experience across FastAPI, Streamlit, and Docker.

EDUCATION

SZABIST Islamabad

Sep 2023 – June 2027

Bachelor of Science in Artificial Intelligence — CGPA: 3.5 / 4.0

Relevant Coursework: Deep Learning, Computer Vision, NLP, Machine Learning, Neural Networks, Data Mining, Database Systems, DSA, Operating Systems, Technical & Business Writing

WORK EXPERIENCE

CodeAlpha — Machine Learning Intern (Remote)

July 2025 (1 month)

- Trained and evaluated supervised regression and tree-based models (Linear Regression, Decision Trees) on structured datasets; improved baseline R^2 and MSE on held-out validation splits through feature engineering and hyperparameter tuning.

PROJECTS

SupportGenie — RAG Customer Support Chatbot | [GitHub](#)

Technologies: Python, FastAPI, LangChain, FAISS, Sentence-Transformers, Llama 3.3 (Groq API), Docker, Hugging Face Spaces

- Built a white-label customer support chatbot that answers questions grounded in an admin-managed knowledge base: document ingestion → chunking → MiniLM embeddings → FAISS retrieval → LLM generation with source citations.
- Engineered a confidence-gated human-handoff mechanism: calibrated the cosine-similarity threshold on on-topic vs. off-topic evaluation queries so the bot escalates to a human instead of hallucinating policies; achieved 83% retrieval hit rate@3 on a held-out evaluation set.
- Designed a provider-agnostic LLM layer (any OpenAI-compatible API — Groq, DeepSeek, OpenRouter) swappable via environment configuration; containerized with Docker and deployed on Hugging Face Spaces.

PixSearch — Content-Based Image Retrieval System | [Live Demo](#) · [GitHub](#)

Technologies: Python, PyTorch, OpenCV, scikit-image, FAISS, Streamlit, Hugging Face Spaces

- Built a visual search engine retrieving the most similar images from a 1,000-image corpus using deep features (ResNet-50, CLIP ViT-B/32) against a classical baseline (HSV histogram + HOG + ORB), achieving mAP 0.93 on retrieval evaluation.
- Indexed L2-normalized embeddings in a FAISS flat inner-product index, achieving sub-millisecond top-K retrieval over the full corpus on CPU.

AgriVision-Pro — Plant Disease Detection System | [GitHub](#)

Technologies: Python, TensorFlow/Keras, OpenCV, Image Processing

- Developed a computer-vision pipeline classifying crop leaf images into healthy vs. multiple disease categories (79% test accuracy) with resizing, normalization, and augmentation-based preprocessing.
- Reported per-class precision and recall alongside overall accuracy to surface minority-class weaknesses rather than relying on a single headline number.

Personal Finance Manager | [GitHub](#)

Technologies: Python, TensorFlow/Keras (LSTM), Pandas, Matplotlib

- Developed a backend-driven financial management application supporting income/expense tracking, budgeting, loans, and recurring payments.
- Integrated LSTM models for expense forecasting (84% accuracy) and built visualizations for cash-flow trends and spending analysis.

CERTIFICATIONS

- Machine Learning with Python — IBM (Coursera): coursera.org/verify/PDLKU2ZZWKBL
- Introduction to Data Science in Python — University of Michigan: coursera.org/verify/WQ9KDVOTF0Z1
- Machine Learning — DeepLearning.AI: coursera.org/verify/JO7EIUHTUB4S

TECHNICAL SKILLS

Languages & Core: Python, C++, SQL, Git, DSA

ML / DL: PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, CNNs, LSTMs, Transfer Learning, OpenCV, scikit-image

GenAI & Retrieval: RAG, LangChain, FAISS, LLM APIs (Groq / OpenAI-compatible)

Deployment: FastAPI, Docker, Streamlit, Hugging Face Spaces